

Pagemark

Complete Documentation

Version 0.4.1 · October 2026

A private, fully local book tracker for Android

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https://github.com/AxialForge/Pagemark · © 2026 AxialForge · MIT License

Pagemark — Package Overview

Version 0.4.1 · for readers outside the project. Three tiers: read the first for the idea, the second for what it does, the third for how it works.

Tier 1 — Simple

What it is. Pagemark is an app for your phone that keeps a diary of the books you read and listen to. For each book you note what you thought, give it a star rating, and track how far you've got. You can look books up by title or by scanning the barcode on the back.

What it's for. Remembering. Which book was that? Did I like it? When did I finish it? It works for any kind of book — fantasy, romance, science, anything — and for audiobooks as well as print.

What makes it different. - It is **private**. Everything stays on your phone. There's no account to make and nobody collects your reading habits. - It has a **hidden shelf** for books you'd rather keep to yourself, locked with a PIN or your fingerprint. - It's **free**, and it uses only free sources for book information.

How you'd use it. Install it, tap *Add book*, scan or search, and start logging. Once a year, glance at the *Stats* tab to see how you did.

Tier 2 — Medium

Purpose. A local-first reading tracker for Android (iOS planned). It replaces cloud services such as Goodreads with a log you fully own, including audiobooks and a private vault.

Main capabilities

Area	What you get
Capture	Search Open Library, look up by ISBN, scan barcodes, add manually; duplicate warning
Logging	Multiple *reads* per book (print / e-book / audio); status, dates, half-star rating, private thoughts, narrator, time
Progress	Current page, progress bars, pages per day, estimated days left
Organise	Shelves (many per book), series with numbers, tags, filters, sort, "Currently reading" strip
Insight	Yearly goal ring with pace forecast, books per month, rating spread, top genres/authors, totals
Privacy	Locked library (PIN with attempt limits and recovery code, or device biometrics), auto-relock, screenshot blocking
Data portability	CSV and JSON export of the main library; whole-database export, optionally passphrase-encrypted
Backup	Scheduled, verified backups to the user's own Google Drive; optional AES-256 encryption; safe restore with automatic safety copy

Area	What you get
Care	Trash with 30-day recovery, in-app updates with release notes, offline version history, diagnostics log
Look	Light/dark × two colour palettes, animated transitions

Principles. No accounts, no analytics, no ads, no paid services. The network is used for exactly three things: book lookup (Open Library), update checks (GitHub Releases), and — only if configured — Drive backup.

Who it's for. Readers who want a durable, private record; people leaving other tracking sites; anyone who wants their data in a plain file they can open elsewhere.

Distribution. Free APKs on GitHub Releases (MIT-licensed source in a public repository). In-app updates install over the old version and keep your data.

Tier 3 — Advanced

Stack

Flutter (Dart) app; **Drift** over **SQLite** for storage; plain `StatefulWidget`s with database streams (no state-management package); `flutter_secure_storage` for secrets; `cryptography` for PBKDF2 and AES-GCM; `workmanager` for scheduled backup; `google_sign_in` + `googleapis` (Drive v3, `drive.appdata` scope) for backup; `http` for Open Library and GitHub; `mobile_scanner` for barcodes; `local_auth` for biometrics.

Data model (schema 3)

- `books` — catalog entry: title, author, ISBN-13, cover URL, genre, series, `seriesNumber`, publisher, year, pages, description, tags, notes, `locked`, `added`, `deletedAt` (soft delete).
- `reads` — one read-through: format, status, started/finished, rating (0.5–5), thoughts, narrator, minutes, `currentPage`; FK to `books` with cascade.
- `shelves`, `book_shelves` — many-to-many.
- Indexes on `reads.bookId`, `books.locked`, `books.title`. WAL journal, 5 s busy timeout, `foreign_keys=ON` in `beforeOpen`. Versioned migrations (1→2→3) covered by a migration test.

Privacy and security design

- **Vault is a flag, not a second database**, enforced at every read path: library queries, Stats, Trash, duplicate check and CSV/JSON export all filter on `locked`. Full DB exports/backups include it by design and can be encrypted.
- **PIN:** PBKDF2-HMAC-SHA256, 150 000 iterations, random 16-byte salt, constant-time compare, persistent attempt counter with exponential lockout (30 s doubling, cap 1 h). One-time recovery code stored only as a PBKDF2 hash. v0.1.0 hashes upgrade on first successful unlock.
- **Vault lifecycle:** on background/leave the app pops locked routes, empties the locked-cover cache and deletes temporary exports; `FLAG_SECURE` is set while the vault is open and kept until resume so the recents thumbnail is blank.
- **Backups:** container `PGMKENC1 | salt | nonce | AES-256-GCM ciphertext | tag`, key from PBKDF2 (150 k). Drive upload is skipped when the plaintext digest is unchanged, size-verified before older copies (newest 5 kept) are pruned. The passphrase lives in secure storage so the background isolate can encrypt unattended.

- **Restore:** open candidate read-only, `PRAGMA integrity_check`, require `books` / `reads`, reject newer schemas, snapshot current DB (newest 3 kept), close connection, delete `-wal/-shm`, rename staged copy into place; app restarts.
- **Android:** `allowBackup=false` plus data-extraction rules exclude everything; no analytics SDKs.

Networking

Open Library search/ISBN/works endpoints with an identifying User-Agent, retry with backoff on 429/5xx/timeouts, per-record fault tolerance, ISBN-10→13 normalisation and checksum validation. Updates use the unauthenticated GitHub releases API: newest-first release list → APK chosen by ABI → streamed download with size and SHA-256 (from the release asset's digest) verification → system installer via `open_filex`. Same signing key ⇒ in-place upgrade.

Quality and process

- 44 automated tests: database behaviour, migration from v1, restore validation, export safety (formula neutralisation, BOM), encryption round-trip and tamper detection, lock service (lockout, recovery, legacy upgrade), lookup parsing/retries (mock HTTP), updater logic, and widget tests for custom UI.
- GitHub Actions CI: analyze, tests, generated-code check, debug build. Signed release APKs are built locally; the keystore never leaves the owner's machine.
- Release flow: version bump + `CHANGELOG.md` in one commit → tag `vX.Y.Z` → APKs (`arm64`, `armv7`) attached to a GitHub Release. The changelog ships inside the app as an asset.

Known limits (0.4.1)

- Database is not encrypted at rest on the phone (SQLCipher is on the roadmap).
- Books deleted from the vault sit in a Trash that doesn't list them, so they can't be restored from the UI.
- Scheduled backups depend on Android's WorkManager timing; OEM battery managers can delay them. No failure notification yet.
- Some plugins add unused storage/notification permissions to the manifest.
- No iOS build yet (project is configured; needs a Mac to build and sign).
- Release builds are not minified (size/safety trade-off pending device testing).

Roadmap highlights

Goodreads/StoryGraph import, grid cover view, quotes and highlights, series entity with total volumes, SQLCipher, backup failure notifications, iOS.

Pagemark — Install Manual

Version 0.4.1 · Android

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1. Requirements

- An Android phone or tablet running **Android 7.0 (API 24) or newer**.
- About **100 MB** free storage (the app is ~30 MB; the library and cached covers are small).
- Optional, for specific features: a camera (barcode scanning), a screen lock or fingerprint (locked library), a Google account (Drive backup), an internet connection (book lookup, updates, backup).

2. Download

Releases are published at **https://github.com/AxialForge/Pagemark/releases**. Open the newest release and pick **one** APK:

File	Use it on
`Pagemark-x.y.z-arm64.apk`	Almost every phone from the last ~8 years (64-bit ARM). **Choose this one** unless it fails to install.
`Pagemark-x.y.z-armv7.apk`	Older 32-bit phones.

Not sure? Download the arm64 file. If Android says the app isn't compatible, use the armv7 one.

3. Install

1. Open the downloaded `.apk` (from your browser's downloads or the Files app).
2. Android will say your browser/Files app isn't allowed to install apps. Tap **Settings** and turn on **Allow from this source**, then go back.
3. Tap **Install**, then **Open**.

Pagemark is signed with the author's key, so future updates install over it and keep your data.

> Play Protect may show "unrecognised app" because Pagemark isn't distributed through the Play Store. Choose **Install anyway** (or **More details** → **Install anyway**).

4. First run

Pagemark opens on an empty Library. Tap **Add book** to start. Allow the **camera** only if you want to scan barcodes; everything else needs no permissions. In **Settings** you can pick a theme and read **About**.

Permissions Pagemark asks for

Permission	Why
Internet	Book lookup (Open Library), update check (GitHub), optional Drive backup
Camera	Scanning ISBN barcodes (asked only when you open the scanner)

Permission	Why
Biometric	Fingerprint/device unlock for the locked library
Install packages	Lets Android offer to install a downloaded update
Background work / network state	Running the scheduled Drive backup

The Android build also lists a few storage/notification permissions that come bundled with libraries it uses; Pagemark does not read your photos or other media. (A later release will strip them from the manifest.)

5. Updating

In the app: Settings → **Updates** → *Download & install*. Android asks once to allow installs from Pagemark. You can also let it check once a day automatically.

By hand: download the newer APK from Releases and open it — it installs over the old version.

Your library, shelves, settings and PIN are all kept. **Make a backup first** if the library matters to you (Settings → Export database file, or Drive backup).

6. Uninstalling — read this first

Uninstalling Pagemark **deletes your library** from the phone (Android's own backup is disabled on purpose). Before you uninstall or switch phones:

1. Settings → **Export database file** (set a passphrase) and keep the file safely, **or** make sure a Google Drive backup is current.
2. On the new phone, install Pagemark and use **Restore** — a backup from an older Pagemark version restores fine; one from a newer version does not.

The PIN is not stored in backups. After restoring, set a new PIN for the locked library.

7. Setting up Google Drive backup (optional, free)

Needs a one-time Google Cloud project because Google requires every app that uses Drive to be registered. Follow `docs/DRIVE_SETUP.md` in the repository (enable the Drive API, create an Android OAuth client with package `com.axialforge.pagemark` and your signing key's SHA-1, create a Web client, copy its Client ID into Pagemark, and publish the consent screen).

For the official release APK, the signing-key SHA-1 is:

`39:04:59:AD:24:0F:35:99:4B:CD:A5:86:81:AF:9C:21:8C:FD:27:F7`

8. Building from source (developers)

`bash git clone https://github.com/AxialForge/Pagemark.git cd Pagemark flutter pub get dart run build_runner build --delete-conflicting-outputs # regenerates lib/db.g.dart flutter analyze && flutter test flutter build apk --release --split-per-abi`

Requirements: Flutter 3.47+ (stable), Android SDK, JDK 17+.

Windows with a very new JDK (25): Gradle can fail with "*Unable to establish loopback connection*". Create a short folder such as `C:\tmpj` and run the build with `JAVA_TOOL_OPTIONS=-Djdk.net.unixdomain.tmpdir=C:/tmpj`.

Signing: create a keystore, put its path and passwords in `android/key.properties` (this file is git-ignored — never commit it). Without it, release builds fall back to the debug key, which can't upgrade a release install.

9. Troubleshooting

Symptom	Fix
"App not installed"	A different signing key is already installed (e.g. a debug build). Back up, uninstall, install the release APK.
"There was a problem parsing the package"	The download was cut off or you picked the wrong architecture; re-download or use the other APK.
Install blocked	Allow "install unknown apps" for the app you opened the APK from.
Update says "No public releases found"	Check connection; the project's releases must be public.
Fingerprint option missing	Set a screen lock/fingerprint in Android settings first.
Scanner is black	Allow the camera permission in Android → Apps → Pagemark.

Pagemark — User Manual

Version 0.4.1 · A private, fully local book tracker for Android

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1. What Pagemark is

Pagemark is a reading log that lives only on your phone. You record what you read or listen to — print, e-book or audiobook, any genre — with your rating, dates, progress and private thoughts. Book details and cover art come from the free **Open Library**; everything you write stays on your device unless you choose to export or back it up.

There is no account, no server and no tracking. The only times Pagemark uses the internet are: looking up a book, checking for an app update, and — if you set it up — backing up to your own Google Drive.

2. The four tabs

Tab	What it is
Library	Your books. Search, filter, sort, shelves, series.
Locked	A private library behind a PIN and/or your fingerprint.
Stats	Your reading goal and charts.
Settings	Themes, updates, export, backup, privacy, trash, diagnostics.

A small dot on **Settings** means an app update is waiting.

3. Adding a book

Tap **Add book** (bottom of the Library).

1. **Search** — type a title or author and press search. Tap a result to add it.
2. **By ISBN** — type or paste an ISBN into the same box and it is looked up directly. Invalid ISBNs are explained instead of searched.
3. **Scan** — tap the barcode icon at the top and point the camera at the ISBN barcode on the back cover. Only real book barcodes are accepted; the torch button helps in dim light.
4. **Manually** — if a book isn't found, choose **Add manually** and enter a title; fill in the rest later with **Edit**.

If the book is already in the same library (same ISBN, or the same title and author) Pagemark warns you and offers to open the existing one. The main and locked libraries never compare against each other, so the main library can't reveal a locked title.

Descriptions are fetched a moment after the book is saved when Open Library has one.

4. The book page

Tap a book to open it. You'll see the cover, details, tags and shelves, your notes, and the **Reading log**.

- **Edit** (pencil) — title, author, series and series number, genre, tags, publisher, ISBN, cover URL, year, pages, description and private notes.

- ⋮ **menu** — move the book to or from the Locked library, or **Move to Trash**.
- **Log a read** — records a new read-through (see §5). A book can have many reads: a re-read, or a print copy and an audiobook of the same title.

Shelves

Tap **+ Shelf** on a book to put it on any number of shelves ("Comfort reads", "Book club", "Lent out"). Create new shelves from there. Remove a book from a shelf by tapping the × on its chip.

Series

In **Edit**, type the series name and the number in the series (decimals work: `2.5`). Rows show the badge, and the **Series** screen (Library ⋮ menu) groups your books, shows how many you've read, and names what's next. It knows only the books you've added, not how long the series really is.

5. Logging a read

Choose **Log a read** (or tap an existing read to edit it).

- **Format** — Print, E-book or Audiobook. Audiobooks add a **Narrator** field.
- **Status** — Want to read, Reading, Finished, or Did not finish.
- Choosing *Finished* fills in today's date (change it if needed); choosing *Reading* or *Want* clears the finished date.
- A finished date earlier than the start date is refused.
- **Started / Finished** — tap to pick a date; **Started today / yesterday** are quick chips.
- **Current page** — shown while *Reading*.
- **Rating** — tap stars. Tap the **left half** of a star for a half star, the right half for a whole one; tap the same value again to clear.
- **Time** — minutes read or listened.
- **Your thoughts** — free text, private.

Delete a read with the bin icon (you'll be asked to confirm).

Reading progress

While a book has an active read, its page shows a **progress card**. Tap **Update progress** and enter the page you're on. If the book has a page count you get a progress bar, pages per day, and an estimate of days left at your pace. Reaching the last page offers to mark the read finished. Progress bars also appear on the Library list and in the **Currently reading** strip.

6. The Library

- **Search** — title, author, tags, genre and series.
- **Filters** — status chips (All / Want / Reading / Finished / DNF), format chips, and one chip per shelf. Tap a shelf chip again to clear it.
- **Sort** (⇅ icon) — Recently added, Title, Author, Rating.
- **Currently reading** — a strip of covers at the top when nothing is filtered.
- **Remembered** — your sort and filters are remembered separately for the main and locked libraries.
- **Ratings** show as stars (4.5 = ★★★★½) and average only your **finished** reads.
- ⋮ **menu** — **Series** and **Manage shelves** (rename or delete shelves; deleting a shelf never deletes books).

7. Stats

The Stats tab counts your **main library only** — locked books never appear in it.

- **Reading goal** — set how many books you want to finish this year. An animated ring shows progress, and Pagemark estimates how many you'll finish by 31 December at your current pace.
- **Finished per month** and **Your ratings** (how your ratings are spread, and your average).
- **Totals** — books tracked, finished, pages, logged time, average days to finish, and finished books by format.
- **Top genres** and **Top authors**.

8. The Locked library

A second library for books you want kept private. Nothing in it shows in the main library, Stats, Trash, exports or searches.

Setting up

Open the **Locked** tab the first time and choose:

- **Set a PIN** — 4 to 12 digits, entered twice. Pagemark then shows a **recovery code** once. **Write it down.** It is the only way back in if you forget your PIN.
- **Use fingerprint / device lock only** — uses your phone's screen lock. Needs a screen lock to be set up on the phone.

Using it

Enter the PIN (or use the fingerprint button). Add books while the Locked tab is open, or move any book across with its ⋮ menu.

Protection you can rely on

- **Wrong PINs are limited.** After 5 misses the PIN is locked out for 30 seconds, doubling after every further 5 misses (up to 1 hour).
- **It re-locks itself** when you leave the tab or put Pagemark in the background. Open locked screens close, locked covers are wiped from the image cache, and temporary export files are deleted.
- **No screenshots.** While the vault is open the screen can't be captured and the app-switcher thumbnail is blank.
- **Changing the lock needs the lock.** Changing or removing the PIN asks for the current PIN (or your fingerprint first).

Forgot your PIN?

On the lock screen tap **Forgot PIN?**, enter the recovery code and a new PIN. You'll receive a **new** recovery code; the old one stops working.

Settings → Privacy

Lock now, **Set or change PIN**, and **Remove lock** (which asks for the current PIN first; locked books stay in the locked library).

9. Themes

Settings → Theme: **Light** or **Dark**, in a **blue** ("Him") or **rose/plum** ("Her") palette. Your choice is remembered.

10. Exporting your data

Settings → Export (main library only — locked books are never included):

- **CSV** — opens in Excel, Numbers or Sheets (accents display correctly; cells that look like formulas are made safe).
- **JSON** — structured, with a format version.

Settings → Database (**includes locked books**):

- **Export database file** — a complete copy you can restore on another device. You'll be asked for a **passphrase**; leave it empty only if you understand the file will be readable by anyone who gets it.
- **Restore from database file** — see §12.

11. Google Drive backup

Settings → **Google Drive backup**. One-time setup (a free Google Cloud project) is described in `docs/DRIVE_SETUP.md`. After that:

1. Paste your **Web client ID** and tap **Sign in**.
2. Turn on **Encrypt backups with a passphrase** (recommended — your locked books are inside every backup). **Remember the passphrase:** without it a backup can't be opened by anyone.
3. Turn on **Automatic backup**, choose **Daily** or **Weekly**, and **Wi-Fi only** if you like.

Backups go to Pagemark's private app-data area on your Drive (they don't appear among your normal files). The **newest 5** are kept. An unchanged library isn't uploaded again, and each upload is verified before older copies are removed. **Back up now** uploads immediately. The Settings page shows the last backup time, or **FAILED** if something went wrong (details are on the backup page).

Android decides exactly when scheduled work runs, so a "daily" backup may be a few hours late, especially with battery saver on.

12. Restoring

From a file (Settings → Restore from database file) or **from Drive** (Backup page → Restore from Drive…):

1. Pick the backup. If it is encrypted you'll be asked for its passphrase.
2. Pagemark checks it is a healthy Pagemark database from this or an older version, and tells you how many books and reads it holds.
3. It keeps a **safety copy of your current library** (the newest 3 are retained inside the app), then replaces the library and asks you to close and reopen the app.

A damaged, foreign, or too-new file is rejected before anything changes.

13. Updates

Settings → **Updates** (or the dot on the Settings tab).

- Shows the version you have and whether a newer one exists, how many releases you're behind, the download size, and **what changed** in each.
- **Download & install** fetches the right file for your phone, checks its size and checksum, and opens Android's installer. The first time, Android asks you to allow installs from Pagemark. The update installs over the current app, so your library and settings are kept.

- **Check automatically** (on by default) does a quiet check at most once a day. The only thing sent is a request to GitHub for the release list. Switch it off any time.
- **Version history** shows every change in every release, offline. After an update, a **What's new** prompt links to it.

14. Trash

Deleting a book moves it to **Settings** → **Trash**. Restore it, or delete it forever; anything left is removed automatically after **30 days**. **Known limitation (0.4.1):** books deleted from the *locked* library also wait 30 days, but the Trash screen deliberately doesn't list locked titles, so they can't be restored from the app — delete a locked book only if you mean it.

15. Diagnostics

Settings → **Diagnostics** shows a small rolling event log (errors, background backup runs, lookup problems). It never contains book titles or your thoughts. **Copy** it when asking for help; **Clear** it when done.

16. Your privacy, in plain terms

- Your library is stored only in a database file inside the app, and Android's own cloud backup is switched off for it.
- Pagemark sends **book searches** to Open Library, **update checks** to GitHub, and — only if you set it up — **backups** to your Google Drive.
- Nothing is sent to the developer. There are no accounts, ads or analytics.
- Cover images are downloaded from Open Library and cached on the phone; locked covers use a separate cache that is cleared whenever the vault closes.

17. Troubleshooting

Problem	What to do
Search says it couldn't reach Open Library	Check your connection and try again; Pagemark retries busy servers automatically. You can always **Add manually**.
A scanned barcode does nothing	Only ISBN book barcodes (starting 978 or 979) are accepted. Try the torch, or type the ISBN.
Covers are blank offline	Covers are downloaded on demand and cached; they appear once they've loaded online.
Forgot the PIN and lost the recovery code	The locked library can't be opened — that is the point of the lock. Removing the lock also needs the PIN, and the PIN is not part of backups, so a restore won't help. Keep the recovery code safe.
Drive sign-in fails	Check the Web client ID, and that the Android OAuth client matches your app's signing key (see DRIVE_SETUP). After changing the client ID, close and reopen Pagemark.
Backup shows FAILED	Open the backup page for the message; also see Settings → Diagnostics.
Update says no releases found	The project's releases page must be public. Check your connection.
"App not installed" during an update	Usually a different signing key (e.g. a debug build over a release build). Back up, uninstall, and install the release APK.

18. Glossary

- **Read** — one read-through of a book (format, dates, rating, thoughts).
- **Shelf** — a free-form group of books; a book can be on many.
- **Locked library** — the PIN-protected second library.
- **Recovery code** — one-time code that resets a forgotten PIN.
- **ISBN** — the number on a book's barcode that identifies an edition.

Pagemark — Release Notes

Every release from the first (0.1.0) to the current one (0.4.1), newest first. Releases are published at https://github.com/AxialForge/Pagemark/releases and the same text is inside the app under Settings → Updates → Version history.

[0.4.1] - 2026-10-12

Fixed

- The book, read and add-book editors now release their text fields when closed (a small memory leak when opening many editors).

Notes

- First release published after the repository went public, so it is the first update the in-app updater can find.

[0.4.0] - 2026-10-12

Day-to-day reading features.

Added

- Reading progress: "Update progress" on the book page stores your current page; a progress bar shows on the list and the "Currently reading" strip, with pages/day and an estimated days-left; reaching the last page offers to mark it finished.
- Yearly reading goal with an animated progress ring and an on-pace forecast (Stats).
- Stats dashboard: finished per month, rating spread, totals (pages, hours, average days to finish), top genres and authors. Locked books never count.
- Shelves: any number per book; filter the library by shelf; manage them from the menu (deleting a shelf keeps the books).
- Series: a series number per book (decimals allowed, e.g. 2.5), series badge on rows, and a Series screen that groups books, counts what you've read and shows what is next.
- "Currently reading" strip at the top of the library.
- Tap-to-rate stars (left half = half star) replace the rating slider; quick "started today/yesterday" chips.
- Sort order and filters (status, format, shelf) are remembered per library.
- Schema 3 (migration from v1/v2 tested): series number, current page, shelves.
- In-app updates (Settings > Updates): checks the project's GitHub Releases, shows how many releases you're behind and what changed in each, downloads the APK for your phone type (arm64 / 32-bit ARM), verifies size and SHA-256, and hands it to Android's installer. It installs over the current app, so your library and settings are kept. A quiet check runs at most once a day (switch it off on the Updates page); Settings shows a dot when an update is waiting.
- "What's new" after an update, and a Version history page - the changelog ships inside the app, so it works offline.

- Your logo throughout the app: launch splash (Android 12+ system splash and older launch screen), Settings header with an About dialog (including the Open Library credit), the Updates page, and friendly empty states on the library and Stats. iOS app icons generated from it.

Changed

- Ratings show as stars (4.5 = ★★★★½) in the theme's accent colour instead of a fixed amber.
- The list leaves room under the last row so the Add button never covers it.
- Stats now update live as you log books.

Fixed

- Bar charts overflowed their box by 1.5 px (caught by the new widget tests).

[0.3.0] - 2026-10-12

Reliability and book-lookup release.

Added

- Diagnostics page (Settings): a small rolling event log (no book titles or thoughts), with Copy and Clear. Uncaught Flutter/async errors and background-backup runs are logged instead of vanishing.
- Typing, pasting or scanning a valid ISBN looks the book up directly; invalid ISBNs are explained instead of searched.
- Book descriptions are now fetched from the Open Library work record (best effort, after the book is saved).
- Barcode scanner: only real book barcodes (valid ISBN, 978/979) are accepted, it returns once, has a torch button, and a hint line. QR codes and price labels are ignored.
- Settings shows backup health (last backup time, or "FAILED") on the Drive backup row.
- GitHub Actions CI: analyze, tests, generated-code check and a debug build on every push.
- 10 more automated tests (lookup parsing, retries, ISBN checksums, log) - 26 total.

Changed

- Lookup is sturdier: identifies itself to Open Library, retries busy/slow responses with backoff, gives readable errors (offline / busy / slow), skips one malformed record instead of failing the whole list, dedupes repeated works, hides non-genre subjects ("Accessible book", "Protected DAISY"), and falls back to searching by ISBN.
- Lists load small cover thumbnails and decode them at display size (less data and memory).
- ISBNs are stored as ISBN-13.
- Changing the Google Web client ID after sign-in now says to restart instead of silently using the old one.
- Replaced deprecated share/sqlite/crypto calls.

[0.2.0] - 2026-10-12

Lock and data-safety release.

Added

- Recovery code for the locked library, shown once when a PIN is set; "Forgot PIN?" resets with it.
- PIN entry is limited: 5 misses lock it out for 30 s, doubling every further 5, capped at 1 h.
- Locked library re-locks when the app goes to the background; its screens close, its cached covers and temporary export files are wiped, and the screen is protected from screenshots / recents thumbnails.
- Passphrase encryption (AES-256-GCM, PBKDF2) for Drive backups and for "Export database file".
- Restore checks the file first (SQLite integrity, Pagemark tables, schema version), keeps a safety copy of the current library (newest 3), and swaps the file in atomically.

- Trash: deleted books are kept 30 days and can be restored (Settings > Trash).
- Duplicate-book warning when adding the same ISBN or title + author (within the same library only).
- Automated tests (16) covering the database, migration, restore validation, export, encryption and lock.

Changed

- Changing or removing the PIN now requires the current PIN (or device auth); a new PIN is entered twice.
- PIN is stored as PBKDF2-HMAC-SHA256 with a random salt (v0.1.0 PINs keep working and upgrade on first unlock).
- "Use device lock only" is remembered.
- Database: foreign keys enforced, WAL + busy timeout, indexes, versioned migration (schema 2).
- Drive backup skips unchanged libraries, verifies the upload size before pruning old copies, and no longer resets the schedule when unrelated settings change.
- Stats ignore locked and trashed books. A book's status comes from its active/latest read; ratings average finished reads only.
- Finished date: setting a book to Finished fills today's date visibly; Reading/Want clears it; finished before started is rejected.
- CSV export neutralises spreadsheet formulas and has a UTF-8 BOM; JSON export carries a format version; export files have timestamps.
- Android auto-backup disabled so the library never reaches Google's cloud backup.

[0.1.0] - 2026-10-11

Added

- Initial release.